

DSA06 – Data Handling and Visualization

Assignment 1: Visualizing Data

Complete Answers with Diagrams – Q1 to Q5

Q1. Visual to R Code: Colour as a Tool to Highlight

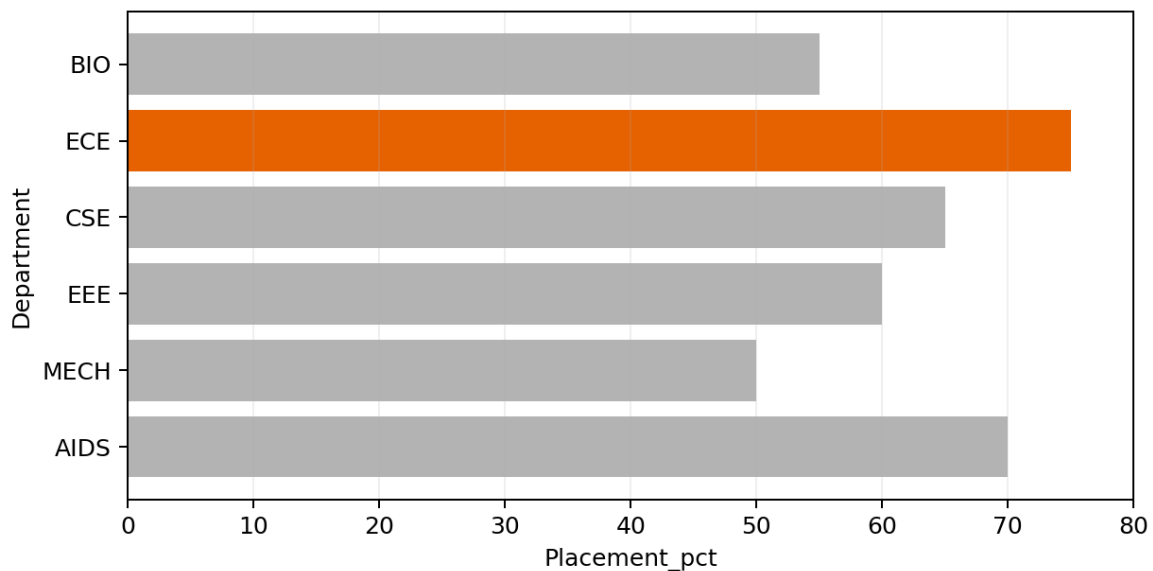
Question:

Write an R program that reproduces the given horizontal bar chart. Use colour to highlight the department with the highest placement percentage. Name the colour technique and explain why the other bars are grey.

Given Data:

Department	Placement pct
BIO	55
ECE	75
CSE	65
EEE	60
MECH	50
AIDS	70

Question Diagram:



Answer:

```
library(ggplot2)
```

```
df <- data.frame(
```

```
Department = c("BIO", "ECE", "CSE", "EEE", "MECH", "AIDS"),
Placement_pct = c(55, 75, 65, 60, 50, 70)
)
```

```
highlight_dept <- "ECE"
```

```
ggplot(df, aes(x = Placement_pct,
               y = reorder(Department, Placement_pct))) +
  geom_col(aes(fill = Department == highlight_dept)) +
  scale_fill_manual(values = c("TRUE" = "#E66100",
                              "FALSE" = "grey70"),
                  guide = "none") +
  labs(x = "Placement_pct", y = "Department") +
  theme_minimal()
```

Colour technique: Highlighting by colour. ECE is orange because it has the highest placement percentage (75%). The other bars are grey to make ECE visually dominant and reduce visual competition.

Q2. Design Decision: Diverging Colour Scales

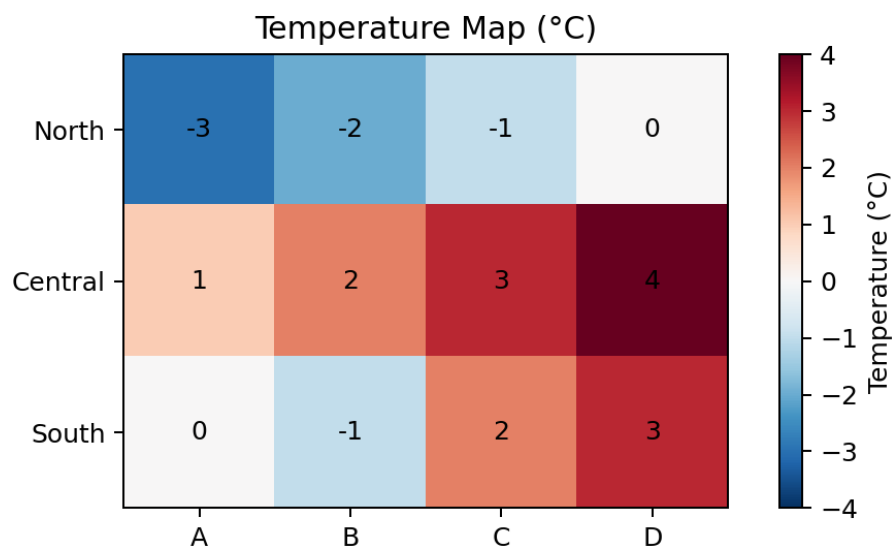
Question:

A city map displays temperature from -3 to +4°C. Decide the appropriate colour scale, midpoint, palette, reason for avoiding rainbow colours, and suitable legend ticks.

Given Data:

Temperature range: -3°C to +4°C; meaningful midpoint = 0°C.

Question Diagram / Example Visualization:



Answer:

A diverging colour scale is appropriate because the data have a meaningful midpoint at 0°C.

Midpoint: 0°C

```
scale_fill_gradient2(  
  low = "blue",  
  mid = "white",  
  high = "red",  
  midpoint = 0  
)
```

Palette: Blue for negative temperatures, white at 0°C, and red for positive temperatures.

Rainbow is a poor choice because it does not give an intuitive ordered numerical progression and some colours are difficult to distinguish for colour-blind viewers.

Suitable legend ticks: -3°C | 0°C | +4°C

Q3. Visual to Table: Reading a Polar Chart

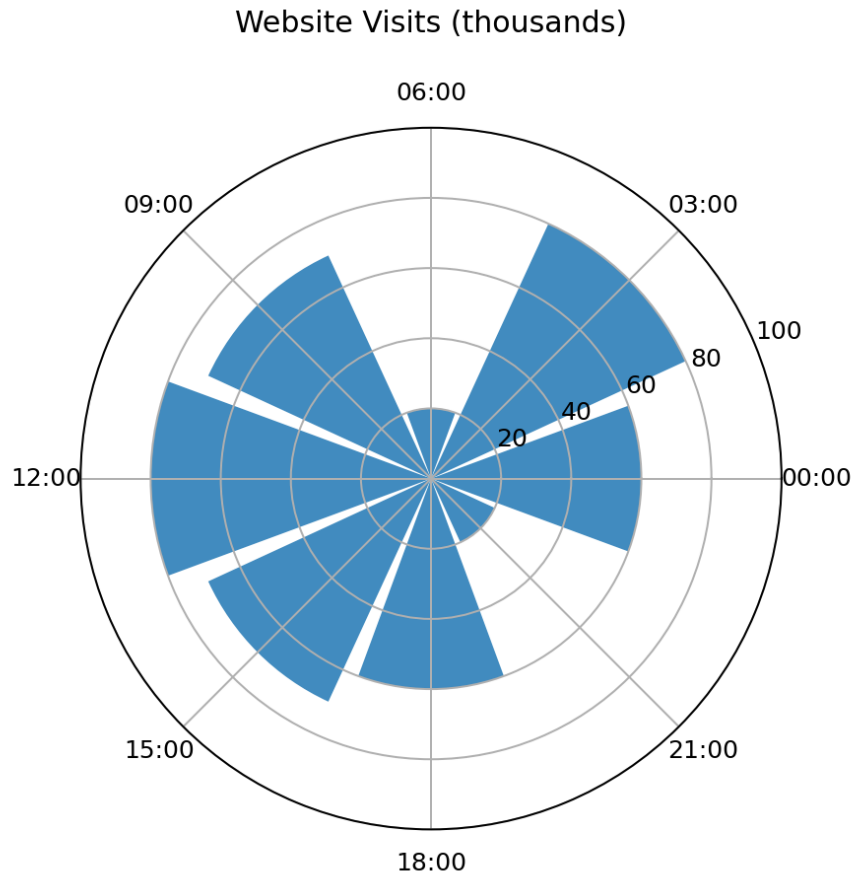
Question:

Reconstruct the table for the polar bar chart showing website visits (thousands) for 8 time slots. Radial grid lines are at 20, 40, 60, 80 and 100. Identify the peak and lowest slots, calculate the angle between adjacent bars, and state one advantage and one drawback.

Given Data:

Number of time slots: 8. Radial scale: 20, 40, 60, 80, 100 thousand visits.

Question Diagram:



Answer – Reconstructed Table:

Time Slot	Web Visits (K)
00:00	60
03:00	80
06:00	20
09:00	70
12:00	80
15:00	70
18:00	60
21:00	20

Angle between adjacent bars: $360^\circ / 8 = 45^\circ$.

Peak: 03:00 and 12:00 – approximately 80 thousand visits.

Lowest: 06:00 and 21:00 – approximately 20 thousand visits.

Advantage: Polar coordinates clearly show cyclic/time-based patterns.

Drawback: Comparing bar lengths around a circle is harder than comparing bars in a normal rectangular chart.

Q4. Table to Visual: Mapping Data onto Aesthetics

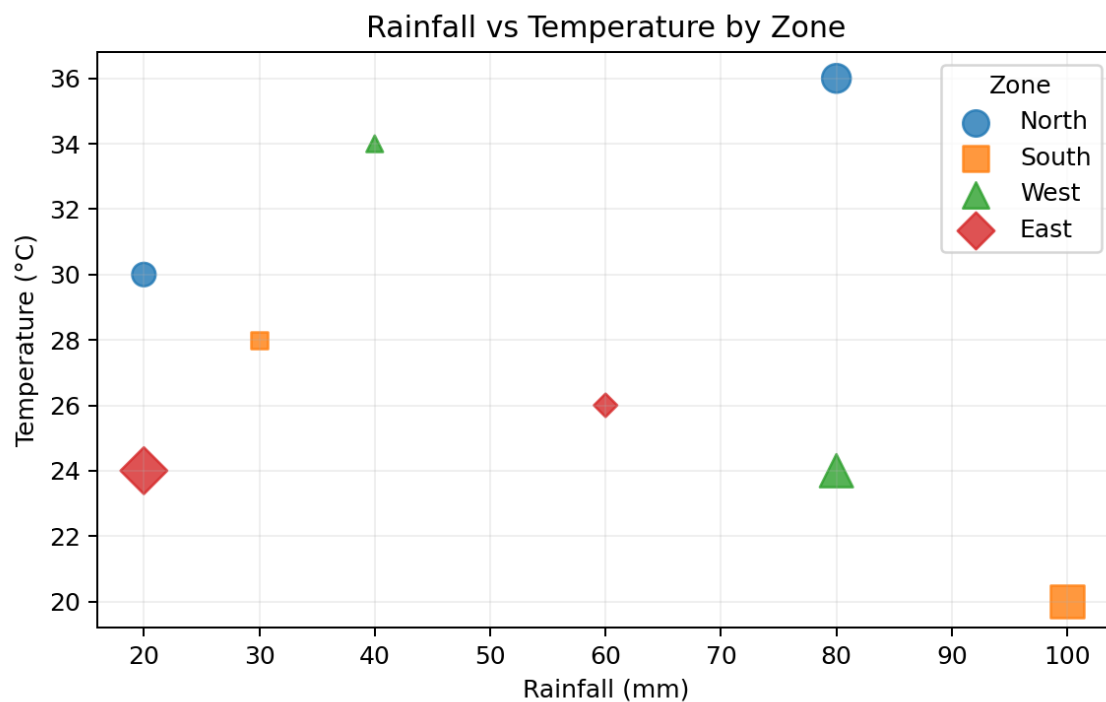
Question:

For the given table, state the data type of each variable, choose suitable aesthetics for a visualization, label the scale types, and explain why Population_M is better mapped to size than shape.

Given Data:

ID	Rainfall_mm	Temp_C	Zone	Population_M
C1	80	36	North	3
C2	100	20	South	4
C3	40	34	West	1
C4	80	24	West	4
C5	30	28	South	1
C6	60	26	East	1
C7	20	30	North	2
C8	20	24	East	4

Question / Answer Diagram:



Answer:

Variable	Data Type
Rainfall mm	Continuous
Temp C	Continuous
Zone	Nominal / Categorical
Population M	Discrete / Quantitative

Suitable aesthetics: X-axis → Rainfall_mm; Y-axis → Temp_C; Colour → Zone; Size → Population_M.

Scale types: Rainfall and Temperature use continuous scales; Zone uses a discrete scale; Population_M uses a quantitative size scale.

```
library(ggplot2)

ggplot(df, aes(x = Rainfall_mm,
               y = Temp_C,
               colour = Zone,
               size = Population_M)) +
  geom_point() +
  labs(x = "Rainfall (mm)",
       y = "Temperature (°C)",
       colour = "Zone",
       size = "Population (M)") +
  theme_minimal()
```

Population_M → shape is a poor choice because shape is mainly suitable for categorical variables. Size communicates numerical magnitude more effectively.

Q5. R Code to Visual: Colour Scale for Data Values

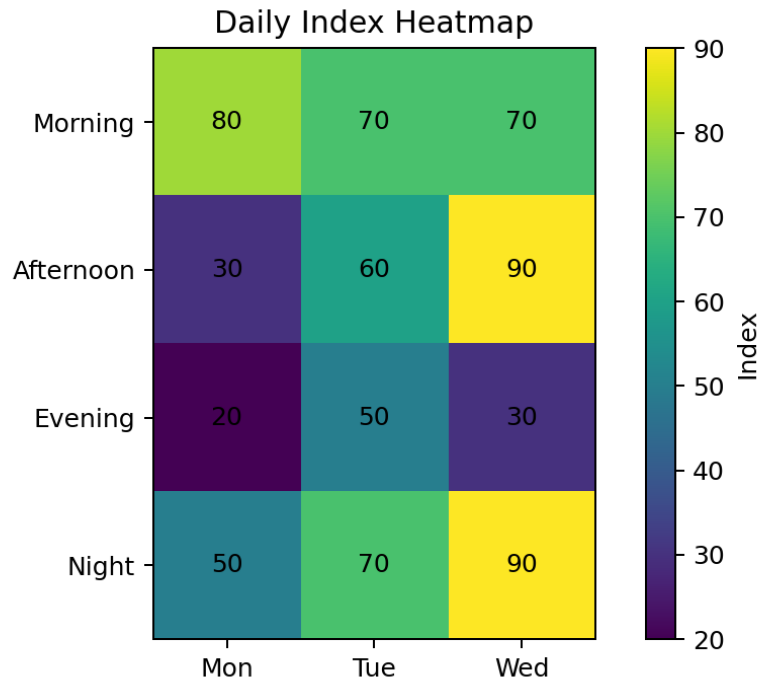
Question:

Read the R code carefully. Sketch the grid produced, identify the dark/light cells and colour bar, state the aesthetic and scale type for Day, Period and Value, and determine whether the scale is sequential or diverging.

Given Data:

Period	Mon	Tue	Wed
Morning	80	70	70
Afternoon	30	60	90
Evening	20	50	30
Night	50	70	90

Question / Answer Diagram:



Answer:

```
ggplot(df, aes(x = Day, y = Period, fill = Value)) +
  geom_tile() +
  scale_fill_viridis_c() +
  labs(fill = "Index")
```

Darkest cells correspond to the lowest values (20 and 30), while the lightest cells correspond to the highest values (90), using the continuous Viridis scale.

Variable	Aesthetic	Scale Type
Day	X-position	Discrete
Period	Y-position	Discrete
Value	Fill / Colour	Continuous

The colour scale is sequential because the values progress from low to high without a meaningful central midpoint. Viridis gives a continuous progression and is suitable for colour-blind viewers.